EDI • MSE

INDUSTRY PROJECT

CLINDERMA

CUSTOMER SUPPORT AI CHATBOT

RAG • PERSONALIZATION • HUMAN HANDOFF

An AI-powered, knowledge-grounded customer support system for a dermatologist-led D2C skincare brand.

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REVIEW SNAPSHOT

Phase 1 + Phase 2 completed
QA benchmark + demo ready

From repetitive support → intelligent support

The goal is not “just a chatbot”. It is a support workflow with grounded AI + escalation.

CUSTOMERS ASK.
SUPPORT TEAMS REPEAT.

Product usage • skincare queries • treatment timelines • orders • support requests

TRADITIONAL FLOW

Customer → WhatsApp / Call → Human → Repeated answer

OUR FLOW

Customer → AI → Verified knowledge → Answer / Human handoff

FAST

CONSISTENT

GROUNDED

HUMAN WHEN NEEDED

The real problem: reliable answers at scale

A support bot must know when to answer — and when NOT to answer.

"How do we automate routine support without creating unsafe or unsupported answers?"

REPETITION

High volume of similar FAQs

HALLUCINATION

LLM may generate unsupported information

CONTEXT

Generic answers may ignore customer history

LANGUAGE

Customers may communicate in English, Hindi or Marathi

ESCALATION

Complex / unresolved queries need a human

WORKFLOW

Orders, leads and assessment flow must coexist

What recent research says

Recent work converges on RAG, domain grounding, contextual relevance and automated evaluation.

PAPER	YEAR	AUTHORS	KEY TAKEAWAY / RELEVANCE
RAGMed: A RAG-Based Medical AI Assistant for Improving Healthcare Delivery	2025	Rajvardhan Patil et al.	Vector retrieval + LLM generation for clinically grounded FAQ responses; directly supports grounded healthcare assistance.
Domain-Aware Healthcare Chatbot Incorporating BERT and RAG	2025	Chunhe Liu et al.	Combines intent classification, LLM generation and RAG to improve factuality, relevance and hallucination control.
Enhancing Healthcare Accessibility: A RAG-Based Medical Chatbot Using Transformer Models	2025	Agrim Kulshreshtha et al.	Uses transformer models + RAG for context-aware healthcare support and reliable responses.
A Multi-module AI System for Intelligent Health Insurance Support Using RAG	2026	Sejal Hanmante et al.	Shows modular RAG, FAISS retrieval and an evaluator feedback loop for quality assessment.

Research sources: MDPI AI 2025; IOS Press / DMI 2025; IEEE ITIKD 2025; Scientific Reports 2026.

What we are building

A website-first support system, designed to extend across channels.

CORE AI

- RAG knowledge retrieval
- Gemini 2.0 Flash
- FAISS vector search
- Grounding / no-answer behavior
- English + Hindi + Marathi

CUSTOMER WORKFLOWS

- FAQ support
- Lead capture
- Kylas CRM flow
- Order tracking
- Human Skin Coach handoff
- Assessment route guard

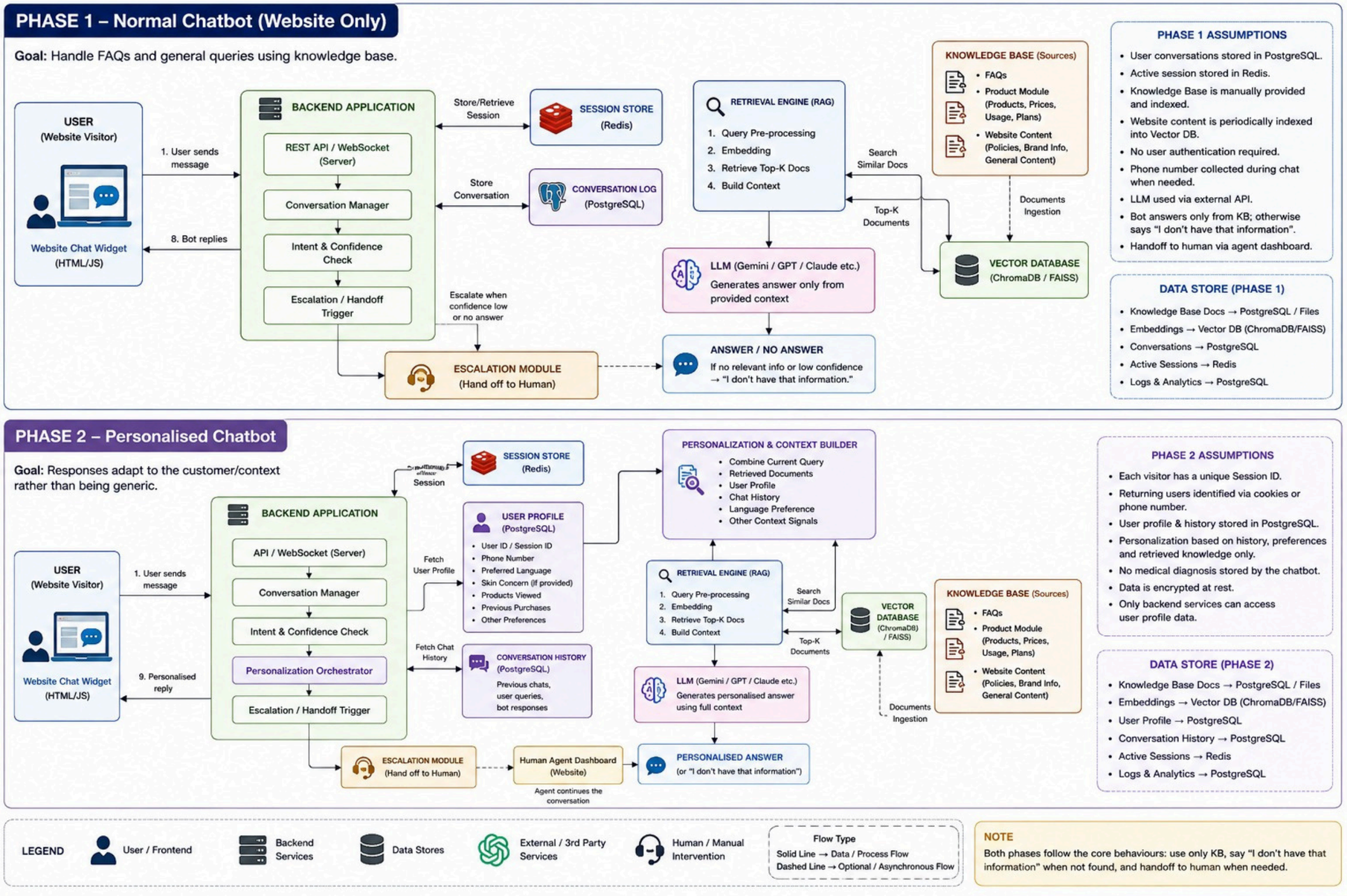
PERSONALIZATION

- User context
- Conversation history
- Preferences
- Context builder
- Personalized answer path

OBJECTIVE **Automate routine support while keeping answers grounded, contextual and safely connected to humans.**

One architecture. Two completed phases.

Phase 1 establishes grounded support. Phase 2 adds customer context.



How one customer query becomes an answer

RAG is the control layer between the user's question and the LLM.

01

QUERY

User sends message

02

EMBED

Convert query to vector

03

RETRIEVE

FAISS → top relevant chunks

04

GROUND

Build context + confidence check

05

GENERATE

Gemini 2.0 Flash

06

RESPOND

Answer OR human handoff

THE SAFETY RULE

If relevant information is not found with sufficient confidence → “I don’t have that specific information” → offer human Skin Coach handoff.

Configured grounding threshold in the project: 0.35

Functional + non-functional

The system is judged not only by what it does, but by how safely and reliably it does it.

FUNCTIONAL REQUIREMENTS

1. FAQ / general query answering
2. RAG-based knowledge retrieval
3. Multilingual responses
4. Personalization using context
5. Lead + phone capture
6. Kylas CRM flow
7. Order tracking
8. Human escalation
9. Skin assessment route guard
10. Agent dashboard

NON-FUNCTIONAL REQUIREMENTS

- Accuracy / grounding
- Reliability
- Security of customer data
- Scalability through pluggable providers
- Maintainability / modular services
- Usability of web widget
- Response-time awareness
- Safe fallback when knowledge is missing

Tech stack

FRONTEND
HTML5 • CSS3 • JavaScript

BACKEND
Python • FastAPI

LLM
Google Gemini 2.0 Flash
GROQ – Fallback

EMBEDDINGS
Gemini Embedding 001

VECTOR SEARCH
FAISS

DATABASE
SQLite

CRM
Kylas

DEV
Python 3.9+ • Browser • Internet

DESIGN CHOICE

Pluggable provider pattern → LLM / vector / CRM / order providers can be swapped through environment configuration.

01 GROUNDED FAQ

“What is purging?”

RAG retrieves the relevant FAQ.

02 PARAPHRASE

“My face is breaking out badly...”

Semantic retrieval finds the matching concept.

03 LANGUAGE

“acne ka treatment kitna time lagta hai?”

Respond in Hindi.

04 ORDER

“Track order CLIN-1001”

Show order status + delivery details.

05 LEAD

“9022905913”

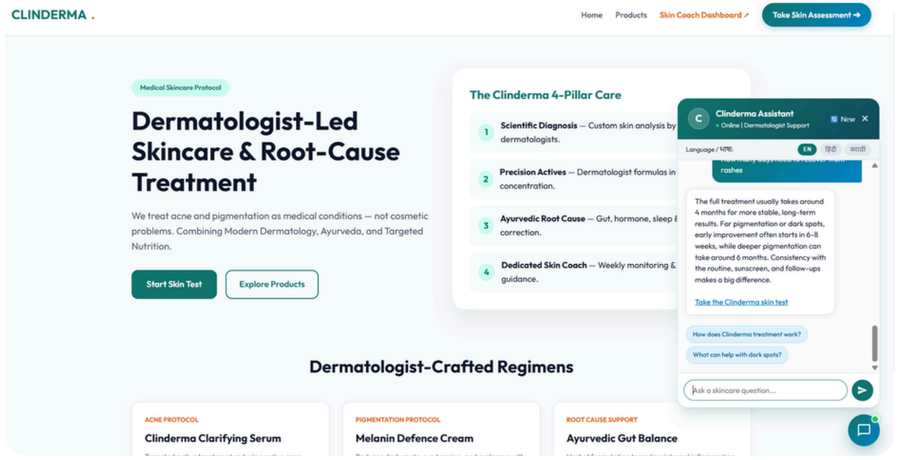
Detect phone → save lead → CRM flow.

06 UNKNOWN

“Capital of France?”

No unsupported answer → handoff path.

DEMO



Who is actually using the system?

Three user groups — one support loop.

CUSTOMER

- Ask FAQs
- Get product / skincare information
- Use English / Hindi / Marathi
- Track orders
- Share phone number
- Request human support

SKIN COACH / AGENT

- Receive escalated sessions
- View chat transcript
- Continue conversation
- Handle complex / unresolved queries
- Work from dashboard

CLINDERMA SUPPORT

- Receive leads
- CRM workflow
- Monitor support interactions
- Maintain knowledge base
- Extend channels over time

We did not stop at “it works.” We measured it.

Automated benchmark across cohort-level FAQ evaluations.

67.7%	74.2%	63.3%	10.7s
OVERALL ALIGNMENT	KEYWORD COVERAGE	KEY-POINT COVERAGE	AVG. LATENCY

COHORT ALIGNMENT



QA TAKEAWAY

32 / 59 evaluations reached ≥70% alignment.

The benchmark also identified low-scoring clinical FAQs for prompt / knowledge improvement.

12 • CONCLUSION

From chatbot → support system

The strongest part of the project is the combination of grounded AI + workflows + evaluation.

01

GROUND

RAG + FAISS + Gemini keeps responses tied to project knowledge.

02

CONTEXT

Phase 2 introduces customer/context-aware response generation.

03

HANDOFF

Unknown / complex queries move to a human Skin Coach.

NEXT

Phase 3: Order Tracking + Call Routing → Phase 4: Final / internal data → Phase 5: Instagram → Phase 6: WhatsApp

“We are not replacing human support — we are making human support more efficient.”

Project + research sources

PROJECT Clinderma Customer Support Chatbot — Project Requirements Document, 2026.

PAPER Patil, R., Abbidi, M., Fannon, S. — “RAGMed: A RAG-Based Medical AI Assistant for Improving Healthcare Delivery.” AI, 2025.

PAPER Liu, C. et al. — “Domain-Aware Healthcare Chatbot Incorporating BERT and RAG.” Frontiers in AI and Applications, 2025. DOI: 10.3233/FAIA250729.

PAPER Kulshreshtha, A. et al. — “Enhancing Healthcare Accessibility: A RAG-Based Medical Chatbot Using Transformer Models.” IEEE ITIKD, 2025. DOI: 10.1109/ITIKD63574.2025.11005179.

PAPER Hanmante, S., Patil, S., Shahade, A.K. — “A Multi-module AI System for Intelligent Health Insurance Support Using Retrieval Augmented Generation.” Scientific Reports, 2026. DOI: 10.1038/s41598-025-31038-6.

THANK YOU